

Common Types of Discrete R.V.'s:

Although discrete r.v.'s are all different, we can group them into "families" sharing common features.

Idea: try to identify different "types" of experiments that we can represent with the same kind of random variable.

Ex. 1:

Population of 100 students

Experiment: Choose a student at random, all are equally likely

$$\rightarrow S = \{1, \dots, 100\}$$

$$\text{Random Variable: } X(\omega) = \begin{cases} 1 & \text{if student } \omega \text{ is in Class of 2028} \\ 0 & \text{o/w} \end{cases}$$

$$\begin{aligned} IP(X=1) &= IP(\{\omega \in S: X(\omega)=1\}) = IP(\{\text{Students in } S \text{ who are Class of 2028}\}) \\ &= \frac{\# \text{ in '28}}{100} = \text{Fraction of '28's} \end{aligned}$$

$$IP(X=0) = IP(\{\omega \in S: X(\omega)=0\}) = IP(\{\text{Students in } S \text{ who are not Class of 2028}\})$$

$$\text{Notice: } \{\omega \in S: X(\omega)=1\} = \{\omega \in S: X(\omega)=0\}^c$$

$$\text{So: } IP(X=1) = 1 - IP(X=0)$$

Hence, the pmf of X has to look like

$$p_X(x) = \begin{cases} \alpha & \text{if } x=1 \\ 1-\alpha & \text{if } x=0 \\ 0 & \text{otherwise} \end{cases}, \text{ for some value of } \alpha \in [0,1]$$

Note that this is not a single distribution/pmf, but rather a family of distributions governed by the parameter α !

Moreover, the value of α tells us something about the population.

Different pmf of $X \Leftrightarrow$ Different $\alpha \Leftrightarrow$ Different population

Ex. 2: DnD combat (simplified)

Experiment: Roll a fair 20-sided die

$$\rightarrow S = \{1, \dots, 20\}$$

Define: $X(\omega) = \begin{cases} 1 & \text{if attack hits, i.e. } \omega > k, \text{ for some } k \in \{2, \dots, 19\} \text{ given} \\ 0 & \text{otherwise} \end{cases}$

$$P(X=0) = P(\{1, \dots, k\}) = \frac{k}{20}$$

$$\rightarrow P(X=1) = 1 - P(X=0) = \frac{20-k}{20}$$

$$\text{So: } P_X(x) = \begin{cases} \frac{20-k}{20} & \text{if } x=1 \\ k/20 & \text{if } x=0 \\ 0 & \text{o/w (otherwise)} \end{cases}$$

Same idea here: different $k \rightsquigarrow$ different distribution of X

Notice that we had two very different experiments, but in both we could describe the distribution of the random variable X in the same way - in terms of x (frac. of '28s') or k (strength of enemy).

This was possible because in both examples, X could only take on 2 possible values: 0 or 1.

$\rightsquigarrow$ So we got the same "family" of distributions for X !

Next:

- 1) Define several such families of discrete r.v.'s - their pdf's and any parameters
 - 2) Describe what kinds of experiments they can represent
 - 3) Study their properties, e.g. Expected Value / Variance
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Bernoulli Distribution:

Def'n: A r.v. X is said to have a **Bernoulli Distribution**

with parameter α , denoted $X \sim \text{Ber}(\alpha)$, if its pmf equals

$$P_X(x; \alpha) = \begin{cases} \alpha & \text{if } x=1 \\ 1-\alpha & \text{if } x=0 \\ 0 & \text{o/w} \end{cases}$$

[Note: We write " $p_X(x; \alpha)$ " to show the pmf's dependence on α .

Parameter: $\alpha \in [0, 1]$

Range: $\{0, 1\}$

Interpretation: $X=1$ means a "success"
 $X=0$ means a "failure"

Parameter α tells us the probability of a "success"

Use: To model any binary outcome. Often, choosing an object, and checking if it satisfies some criterion. Examples of "Success":

- Winning a (potentially unfair) coin flip; $\alpha = \text{prob. of winning}$
- A student in class being an ORF major; $\alpha = \text{frac. that is ORF}$
(Assuming a single student is selected at random, w/ equally likely outcomes)

Properties: If $X \sim \text{Ber}(\alpha)$, then

- $E[X] = \alpha$
- $\text{Var}(X) = \alpha(1-\alpha)$

Note: If $\alpha=0$ or $\alpha=1$, then X isn't random!

Hence, its Variance should be zero.

Geometric Distribution

Def'n: We say $X \sim \text{Geo}(\alpha)$, i.e. X has a geometric dist'n with parameter α , if its pmf equals:

$$P_X(x; \alpha) = (1-\alpha)^{x-1} \cdot \alpha, \text{ for } x=1, 2, 3, \dots$$

It's cdf is given by:

$$F_X(x; \alpha) = 1 - (1-\alpha)^{\lfloor x \rfloor} \text{ for } x \geq 1, \text{ where } \lfloor x \rfloor = \text{"largest integer } \leq x"$$

$\hookrightarrow$ read: "floor of x "

Parameter: $\alpha \in (0, 1]$

Range: $\{1, 2, 3, \dots\}$

Interpretation/Use: To model the number of trials required to obtain the first success, where trials:

i) all have probability of success given by α

ii) are mutually independent [Note: We haven't formalized this yet... But soon!]

Think: $A_1 = \{\text{1st fails}\}, A_2 = \{\text{2nd fails}\}, \dots, A_k = \{\text{k-th fails}\}, A_{k+1} = \{\text{k+1-th succeeds}\},$

then the events $A_1, A_2, \dots, A_k, A_{k+1}$ are mutually independent.

$$\begin{aligned} \Rightarrow \underbrace{P(F, F, F, \dots, F, S)}_{\substack{k \text{ failures, } k+1 \text{ total trials}}} &= P(A_1 \cap A_2 \cap \dots \cap A_k \cap A_{k+1}) = \underbrace{P(A_1)}_{1-\alpha} \cdot \dots \cdot \underbrace{P(A_k)}_{1-\alpha} \cdot \underbrace{P(A_{k+1})}_{\alpha} \\ &= (1-\alpha)^k \cdot \alpha = \underline{P_X(k+1; \alpha)} \end{aligned}$$

I.e. It is a waiting time.

Properties: If $X \sim \text{Geo}(\alpha)$, then

$$\bullet E[X] = \frac{1}{\alpha}$$

$$\bullet \text{Var}(X) = \frac{1-\alpha}{\alpha^2}$$

Ex: "Trial": Shuffling a deck of cards, revealing top card (without removing it)

"Success": Whenever the revealed card is an Ace

$\leadsto$ # of shuffles needed to reveal the first Ace is $\text{Geo}(1/13)$!

Binomial Distribution:

Def'n: We say $X \sim \text{Bin}(n, \alpha)$, i.e. X has a binomial distribution with parameters n, α , if its pmf is:

$$p_x(x; n, \alpha) = \binom{n}{x} \alpha^x (1-\alpha)^{n-x}, \text{ for } x = 0, 1, 2, \dots, n$$

Parameters: $\alpha \in [0, 1]$, $n = 1, 2, 3, 4, \dots$

Range: $\{0, 1, 2, \dots, n\}$

Interpretation/Use: To model the # of successes in n trials, where the trials: i) are mutually independent, and ii) all have success probability of α .

Properties: If $X \sim \text{Bin}(n, \alpha)$, then

- $E[X] = n \cdot \alpha$
- $\text{Var}(X) = n \cdot \alpha(1-\alpha)$
- If $Z_1, Z_2, \dots, Z_n$ are independent $\text{Ber}(\alpha)$, then $Z_1 + Z_2 + \dots + Z_n \sim \text{Bin}(n, \alpha)$

Note: Why does the pmf include $\binom{n}{x}$, i.e. $C(n, x)$?

Ans: the order of the trials does not matter! There are $\binom{n}{x}$ different sequences of length n that contain exactly x successes and $n-x$ failures. Each such sequence occurs with prob. $\alpha^x \cdot (1-\alpha)^{n-x}$

$\leadsto$ So, to find the prob. of any sequence with exactly x successes, we have to add the probabilities of each!

$$\Rightarrow P(x \text{ successes from } n) = \underbrace{\alpha^x \cdot (1-\alpha)^{n-x} + \alpha^x \cdot (1-\alpha)^{n-x} + \dots + \alpha^x \cdot (1-\alpha)^{n-x}}_{\binom{n}{x} \text{ identical terms}}$$

Ex: Prob. of a sequence for $n=3$, $x=2$

$$P(S, S, F) = \underbrace{P(1^{\text{st}} S)}_{=\alpha} \underbrace{P(2^{\text{nd}} S)}_{=\alpha} \underbrace{P(3^{\text{rd}} F)}_{=1-\alpha}, \text{ by mutual indep.}$$

$$\leadsto P(SSF) = \underline{\underline{\alpha^2(1-\alpha)}}$$

Poisson Distribution:

Def'n: We say $X \sim \text{Poi}(\lambda)$ if its pmf equals:

$$P_X(x; \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}, \text{ for } x = 0, 1, 2, \dots$$

Parameter: $\lambda > 0$

Range: $\{0, 1, 2, \dots\}$

Interpretation/Use: To model the # of times an event occurs within a fixed window of time, such that:

i) events occur independently

ii) the avg. rate at which events occur $(= \frac{\lambda}{\text{time window}})$ is constant

Examples:

- # of times the Nile floods in 10 years
- # of times the Ohio floods in 10 years
- # of vehicles passing a toll on I-95 between 8am-10am

↳ A shared feature of Poisson R.V.'s

Properties: If $X \sim \text{Poi}(\lambda)$

- $E[X] = \lambda$
- $\text{Var}(X) = \lambda$
- If $X \sim \text{Poi}(\lambda_x)$ and $Y \sim \text{Poi}(\lambda_y)$ are independent, then $X + Y \sim \text{Poi}(\lambda_x + \lambda_y)$

Intuition: Example where $\lambda_x = \lambda_y$.

Imagine we have a time window of 1 hr. Compare:

- Adding the # of occurrences within two 1-hr periods
- Model the # of occurrences in a 2-hr window, but

keeping the avg. rate of occurrence constant.

$\Rightarrow$ Param. of Poisson r.v. must double!

Hence: $\text{Poi}(\lambda) + \text{Poi}(\lambda) = \text{Poi}(2\lambda)$

Continuous R.V.'s:

Many quantities can take on an uncountably infinite # of values.

E.g.:

- Inches of snowfall
 - Duration of a call
 - Weight of a panda
- } Can be any non-neg. real number

If a random variable has an uncountably inf. range, it is called **continuous**.

To model these as r.v.'s, what is the prob. they equal a specific exact value?

E.g.: How likely am I to get 3.1415... inches of snow?

→ Zero! Why? Because an exact number (here, π) is too precise

An Example:

Say we want a r.v. X to have range of $[0, 1]$, and each value in its range is equally likely.

Goal: Show that $IP(X=x) = c > 0$, for all $x \in [0, 1]$ cannot occur.

Consider the following events:

$$A_1 = \{X = 1\}, A_2 = \{X = \frac{1}{2}\}, \dots, A_k = \{X = \frac{1}{k}\}, \dots$$

Notice:

• $A_1, A_2, \dots$ are mutually exclusive $\Rightarrow$ Can use Axiom 3

• Their union is a strict subset of $[0, 1]$. (Recall: $U_{i=1}^{\infty} A_i \subset [0, 1]$, but $U_{i=1}^{\infty} A_i \neq [0, 1]$)

$$\Rightarrow IP(U_{i=1}^{\infty} A_i) \leq IP(S) = 1$$

By Axiom 3:

$$\begin{aligned} IP(U_{i=1}^{\infty} A_i) &= \sum_{i=1}^{\infty} IP(A_i) \\ &= \sum_{i=1}^{\infty} c \quad \text{= } c \text{ by assumption} \end{aligned}$$

$$IP(U_{i=1}^{\infty} A_i) = \infty? \quad \swarrow c > 0$$

→ Should be at most 1... $\Rightarrow$ We must have $c = 0!$